



FACULTY OF MATHEMATICS,  
PHYSICS AND INFORMATICS

Comenius University  
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3D Vision

# Lecture 9: Monocular Depth Estimation and 3D Object Detection

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- Monocular Depth Estimation
- Pose/Geometry Estimation
- Novel View Synthesis

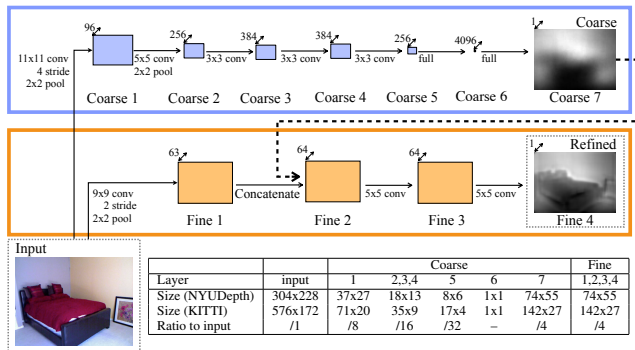


The goal of monocular depth estimation is to take a single RGB image and output a depth image for the given scene. This task lends itself very well to deep learning based approaches as the geometry can be inferred from learned structures. We will briefly cover some interesting approaches to this problem.

The most significant way in which we can divide the methods is based on whether they need depth annotations or not. Based on this we can divide them into

- Supervised methods - require pairs of corresponding RGB and depth data
- Semi-supervised methods - require some other information - usually two stereo RGB images of the same scene
- Unsupervised methods - usually utilize sequences of RGB images

# Simple Supervised Approach



$$L(y, y^*) = \frac{1}{n} \sum_i y_i^2 - \frac{\lambda}{n^2} \left( \sum_i y_i \right)^2 \quad y_i = \log d_i - \log d_i^* \quad (1)$$



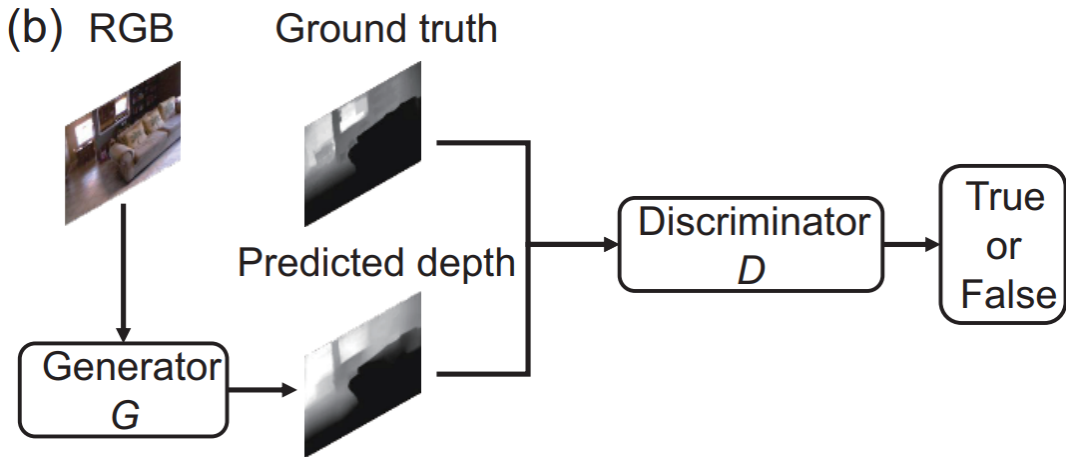


Image adopted from: Chaoqiang Zhao et al. "Monocular depth estimation based on deep learning: An overview." In: *Science China Technological Sciences* 63.9 (2020), pp. 1612–1627

# Semi-supervised Training

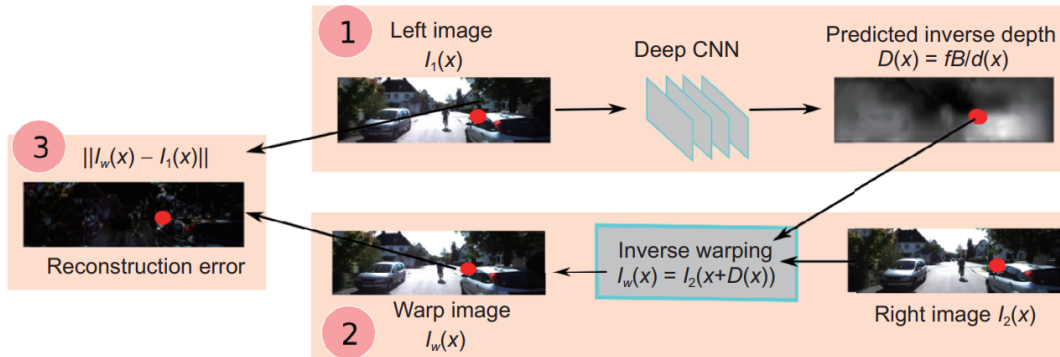


Image adopted from: Chaoqiang Zhao et al. "Monocular depth estimation based on deep learning: An overview." In: *Science China Technological Sciences* 63.9 (2020), pp. 1612–1627

# Sparsely- and Semi-supervised Training

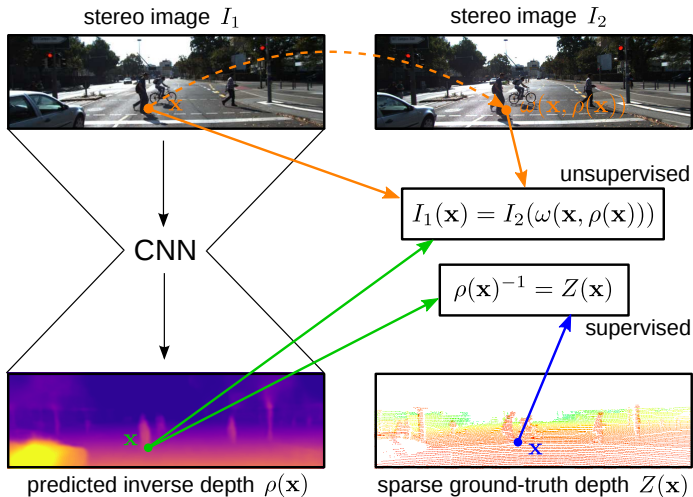


Image adopted from: Yevhen Kuznetsov, Jorg Stuckler, and Bastian Leibe. "Semi-supervised deep learning for monocular depth map prediction." In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2017, pp. 6647–6655

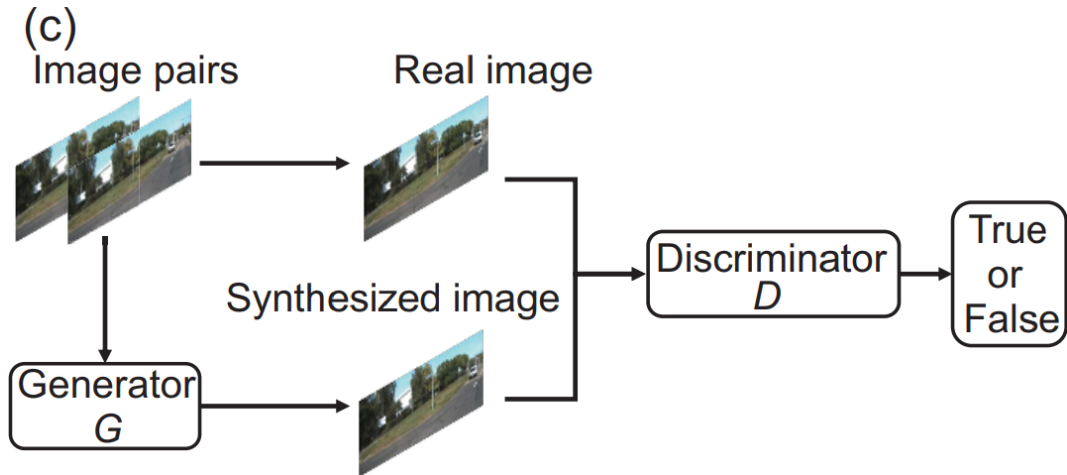


Image adopted from: Chaoqiang Zhao et al. "Monocular depth estimation based on deep learning: An overview." In: *Science China Technological Sciences* 63.9 (2020), pp. 1612–1627

# Unsupervised Training

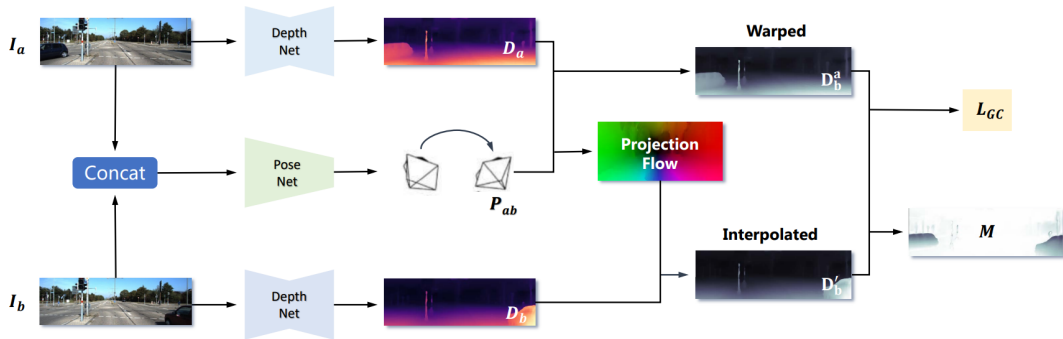


Image adopted from: Jiawang Bian et al. "Unsupervised scale-consistent depth and ego-motion learning from monocular video." In: *Advances in neural information processing systems* 32 (2019)



Problem: Early datasets had various limitations:

- LIDAR - accurate, but sparse
- RGB-D cameras - accurate but limited distance (viable for indoor only)
- Stereo disparity - depends on method used
- SfM - depth known only up to scale

Solution - different types of depth prediction:

- Affine invariant - depth known up to unknown global scale and shift
  - ▶  $d_i = s \cdot p_i + u$
- Scale invariant - depth known up to unknown global scale
  - ▶  $d_i = s \cdot p_i$
- Metric depth - estimated directly in m or mm.

# Depth - focal ambiguity

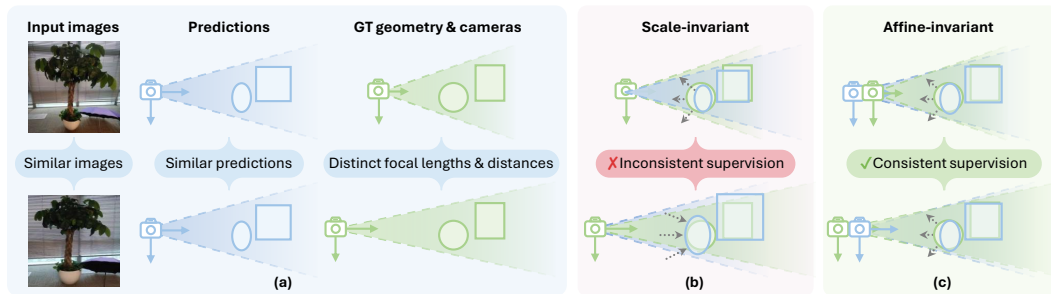


Image adopted from: Ruicheng Wang et al. "Moge: Unlocking accurate monocular geometry estimation for open-domain images with optimal training supervision." In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2025, pp. 5261–5271

# How to Train an Affine-invariant Depth



MoGe<sup>5</sup> trains a network to predict affine-invariant point map  $\mathbf{P} \in \mathbb{R}^{H \times W \times 3}$  with loss:

$$\mathcal{L}_G = \sum_{i \in \mathcal{M}} \frac{1}{z_i} \|s\hat{\mathbf{p}}_i + \mathbf{t} - \mathbf{p}_i\|_1, \quad (2)$$

with optimal scale and shift:

$$(s^*, \mathbf{t}^*) = \operatorname{argmin}_{s, \mathbf{t}} \sum_{i \in \mathcal{M}} \frac{1}{z_i} \|s\hat{\mathbf{p}}_i + \mathbf{t} - \mathbf{p}_i\|_1, \quad (3)$$

Computing (3) is non-trivial. Previous works used some approximations. MoGe transforms the problem into a series of parallel 1-D optimization problems.

<sup>5</sup>Ruicheng Wang et al. “Moge: Unlocking accurate monocular geometry estimation for open-domain images with optimal training supervision.” In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2025, pp. 5261–5271





The affine-invariant point map representation can be used to recover both camera shift and focal length. Given the predicted 3D points  $(x_i, y_i, z_i)$  and their corresponding 2D pixels  $(u_i, v_i)$ , we solve for camera focal length prediction  $f$  and Z-axis shift  $t'_z$  by minimizing the projection error:

$$\min_{f, t'_z} \sum_{i=1}^N \left( \frac{fx_i}{z_i + t'_z} - u_i \right)^2 + \left( \frac{fy_i}{z_i + t'_z} - v_i \right)^2, \quad (4)$$

where  $t'_z = t_z/s$ .

Scale can additionally be predicted by a network trained using the optimal scale  $s^*$  from previous slide.<sup>6</sup>

<sup>6</sup>Ruicheng Wang et al. "Moge-2: Accurate monocular geometry with metric scale and sharp details." In: *Advances in Neural Information Processing Systems* 38 (2026), pp. 35928–35959

# Recovering Focal Length

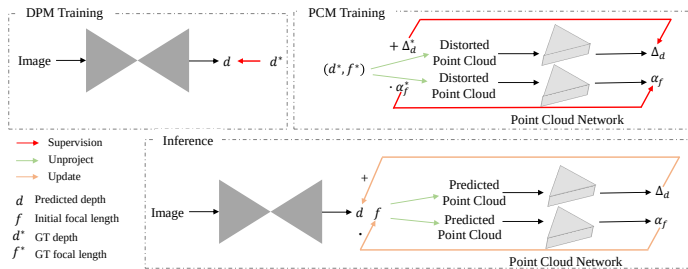
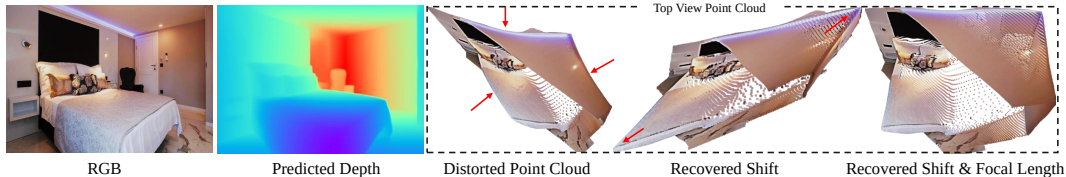


Image adopted from: Wei Yin et al. "Learning to recover 3d scene shape from a single image." In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2021, pp. 204–213

# Deep F without Correspondences

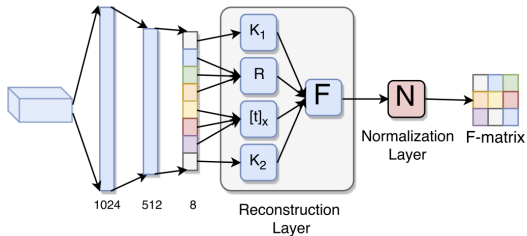
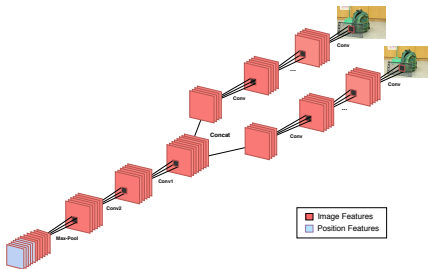
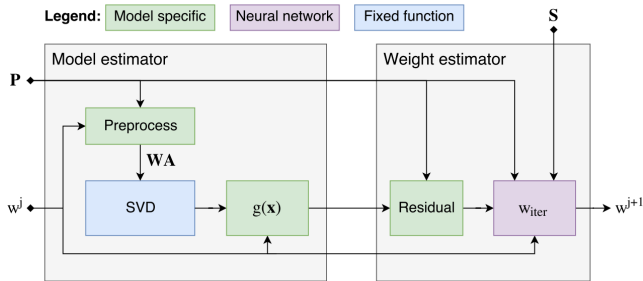


Image adopted from: Omid Poursaeed et al. "Deep fundamental matrix estimation without correspondences." In: *Proceedings of the European Conference on Computer Vision (ECCV) Workshops*. 2018, pp. 0–0

# Deep F Estimation

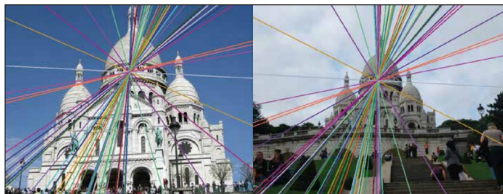


The network uses as inputs point correspondences and learns to obtain weights for iterative least squares loop.

# Learning to find good models in RANSAC



Model hypothesis  $h$



Error prediction from residuals by network  $f$

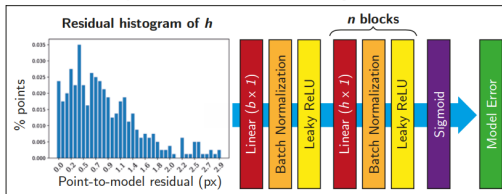


Image adopted from: Daniel Barath, Luca Cavalli, and Marc Pollefeys. "Learning to find good models in RANSAC." In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2022, pp. 15744–15753

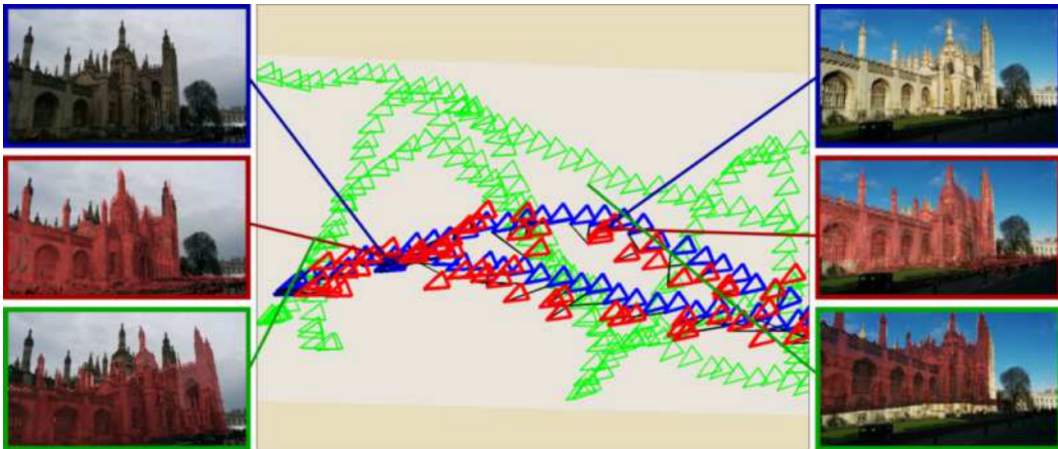


Image adopted from: Alex Kendall, Matthew Grimes, and Roberto Cipolla. "Posenet: A convolutional network for real-time 6-dof camera relocalization." In: *Proceedings of the IEEE international conference on computer vision*. 2015, pp. 2938–2946

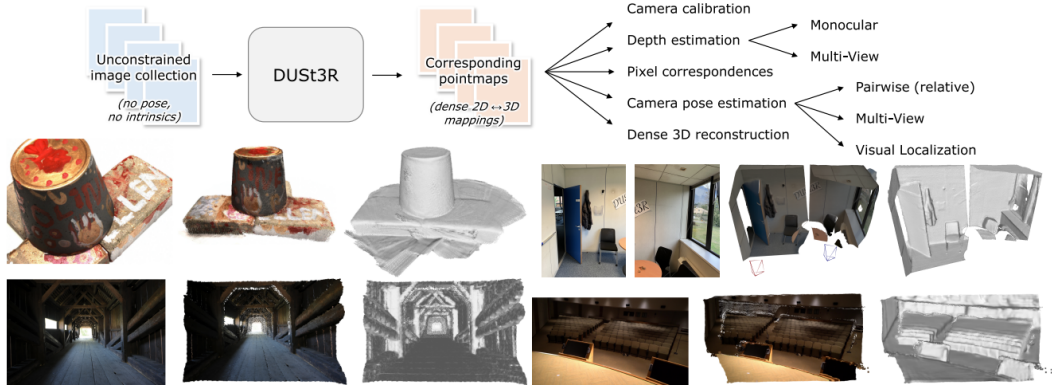


Image adopted from: Shuzhe Wang et al. "Dust3r: Geometric 3d vision made easy." In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024, pp. 20697–20709

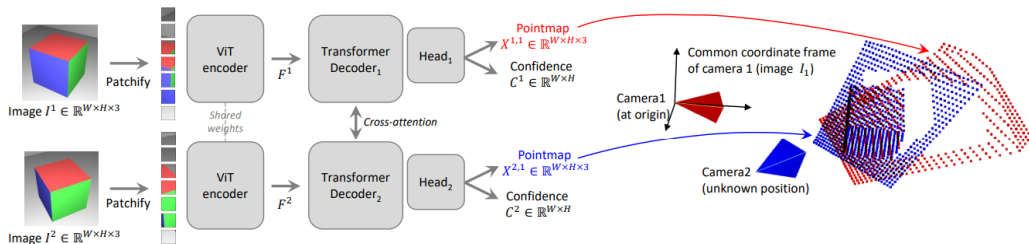


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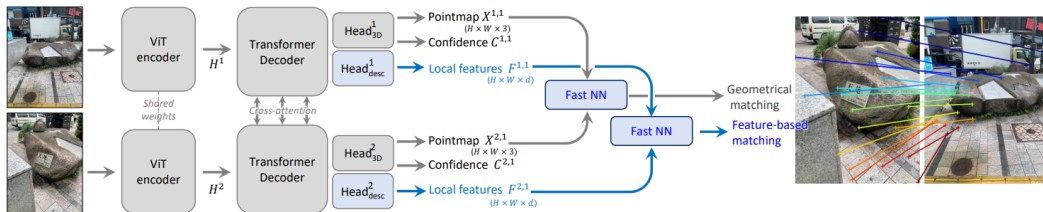


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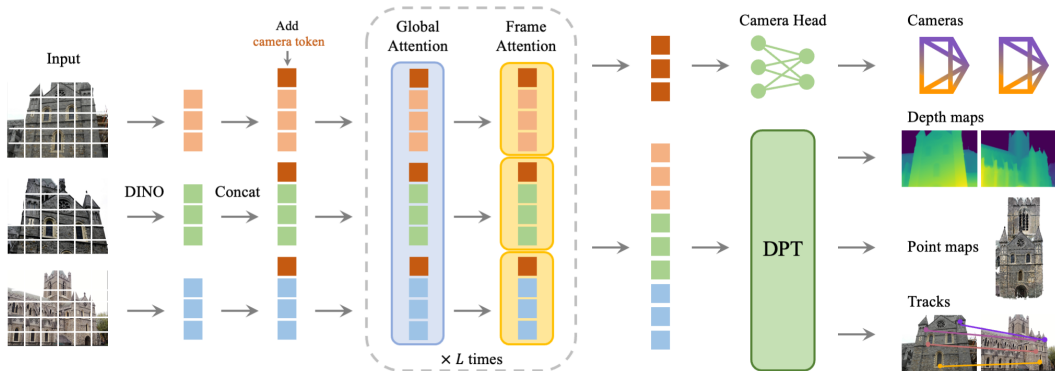
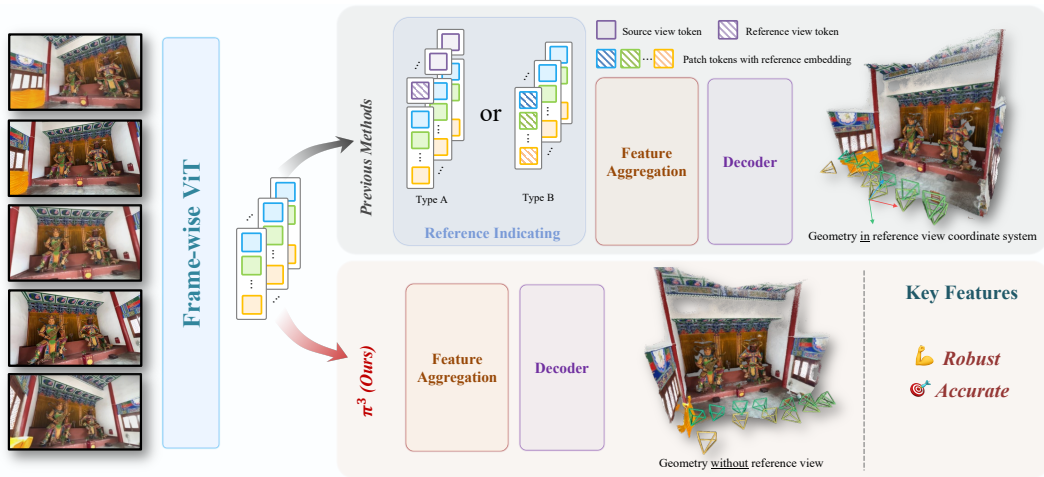


Image adopted from: Jianyuan Wang et al. "Vggt: Visual geometry grounded transformer." In: *Proceedings of the Computer Vision and Pattern Recognition Conference*. 2025, pp. 5294–5306

# Permutation invariance



# Neural Radiance Fields

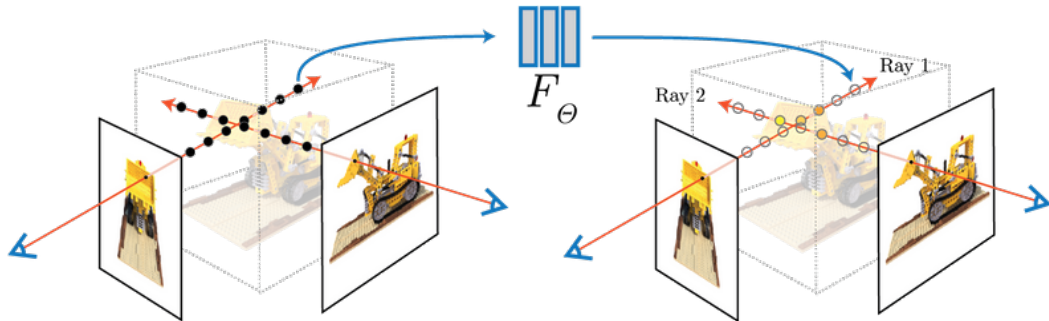


Image adopted from: Ben Mildenhall et al. "Nerf: Representing scenes as neural radiance fields for view synthesis." In: *Communications of the ACM* 65.1 (2021), pp. 99–106

# Neural Radiance Fields

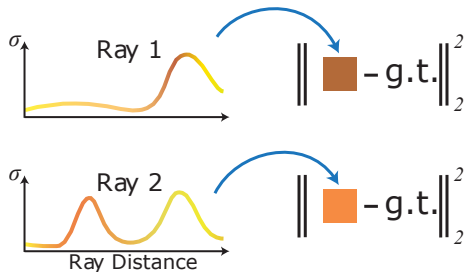
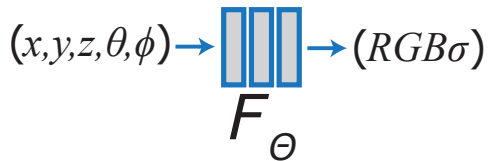


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# Inverted NeRF

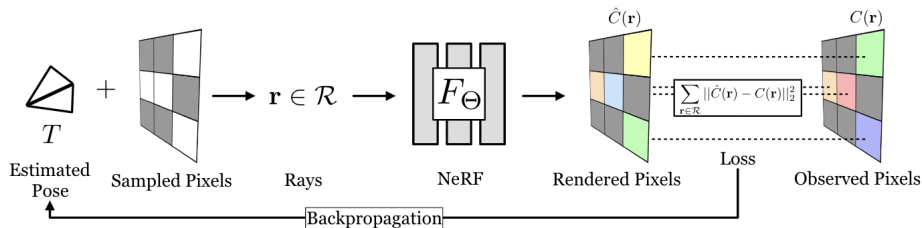


Image adopted from: Lin Yen-Chen et al. "iNeRF: Inverting Neural Radiance Fields for Pose Estimation." In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2021

# Gaussian Splatting

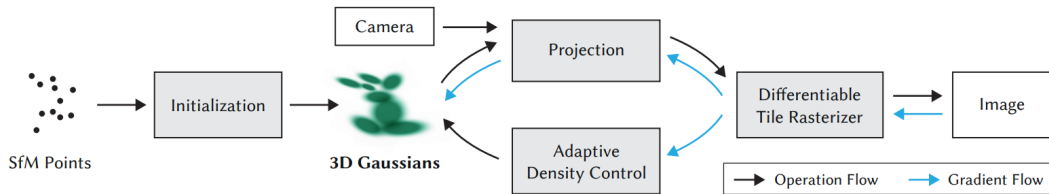


Image adopted from: Bernhard Kerbl et al. "3d gaussian splatting for real-time radiance field rendering.." In: *ACM Trans. Graph.* 42.4 (2023), pp. 139–1

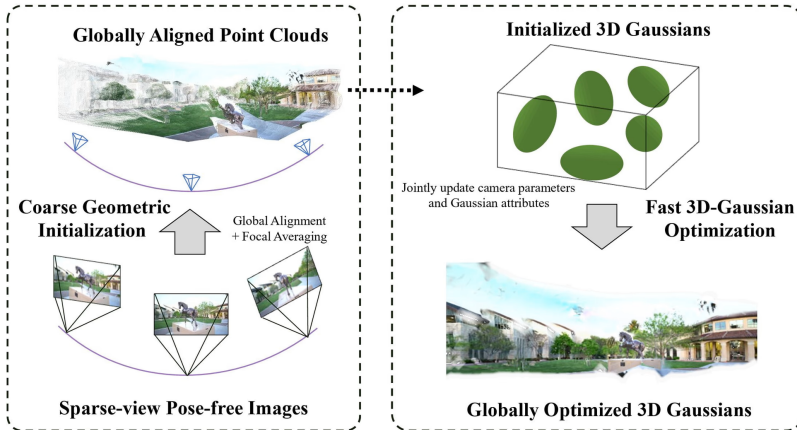


Image adopted from: Zhiwen Fan et al. "InstantSplat: Unbounded sparse-view pose-free gaussian splatting in 40 seconds." In: *arXiv preprint arXiv:2403.20309* 2.3 (2024), p. 4